MILITARY SPECIFICATION

PACKAGING AND PROCESSING OF FOODS IN COLLAPSIBLE CONTAINERS (TUBES)

This specification is approved for use by the U.S. Army Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the packaging and processing of semi-solid and liquid foods in collapsible tubes. The products are a component of operational rations for the Department of Defense.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specifications For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Components and materials.

3.2.1 Food product. The food product to be filled, sealed, and processed shall be as specified in the applicable product document (see 6.1).

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3.2.2 Tube material. The tubes shall be made from aluminum slugs with a purity of 99.7 percent (%) aluminum.

3.2.3 Tube construction. The tubes shall have a length of 6-3/4 inches ± 1/32 inch and a diameter of 1.495 inches minimum to 1.505 inches maximum. The tube wall thickness shall be 0.005 inch minimum and 0.0065 inch maximum. One end of each tube shall be fitted with a No. 28 blind opening with a 0.004 ± 0.002 inch thick diaphragm recessed 0.100 ± 0.016 inch. The neck shall project 0.312 ± 0.016 inch from the shoulder and shall be provided with a 60° -20 turns per inch (T.P.I.) thread. Figure 1 illustrates the tube dimensions. The tubes shall be coated overall on the inside with internal lacquer H23, epoxy phenolic linear. The tubes shall be free from metal exposure when tested in accordance with 4.6.1. When specified (see 6.1), the exterior of each tube body shall be coated overall with a coating resistant to processing in accordance with TT-C-495.

3.3 Tube filling and sealing. The tubes shall be filled and sealed as specified in the product document. The filled tubes shall be immediately given a saddle-back fold and crimped securely, then sealed under a vacuum by flushing the head space of each filled tube with super-heated steam. The fold on each side of the saddle-back closure shall be approximately 5 millimeters in length.

3.4 Thermoprocessing. The filled and sealed tubes shall be processed within two hours after sealing, as specified in 3.4.1 or 3.4.2, in accordance with the requirements of the product document.

3.4.1 Retorting. The filled and sealed tubes shall be thermostabilized by retorting at a temperature of 240°F or below until a sterilization value (F_o) of not less than 6 has been achieved. The product shall be immediately cooled to 100°F or lower. A plastic cap shall be placed on the threaded end of each cooled tube. The filled, sealed, and thermoprocessed tubes shall show no evidence of tube swelling when tested as specified in 4.5.4.

3.4.2 Pasteurization. The filled and sealed tubes shall be pasteurized by immersion in a hot water bath. Each particle of product shall receive a temperature of not less than 190°F for 10 minutes. The product shall be immediately cooled to 100°F or lower. A plastic cap shall be placed on the threaded end of each cooled tube. The filled, sealed, and pasteurized tubes shall be float tested as specified in 4.5.5. The filled, sealed, and pasteurized tubes shall show no evidence of tube swelling when tested as specified in 4.5.4.

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3.5 Unit packing of filled tubes. After the filled tubes have reached room temperature, seventy-two filled tubes, arranged nine by eight, shall be unit packed on end with orifices on the bottom in a paperboard box conforming to style III, type C, class j, subclass 1 of PPP-B-566. The dimensions of the box shall be 14-5/8 inches in length by 13 inches in width by 8 inches in depth (± 1/8 inch). In addition, tubes shall be separated by a full height paperboard partition.

3.6 Packing for shipment. Seventy-two tubes of product, unit packed as specified in 3.5, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic, grade 275 of PPP-B-636. Shipping containers shall be closed in accordance with method II of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

3.7 Labeling and marking.

3.7.1 Tubes. After processing and cooling, the filled tubes shall be dried and shall be provided with a paper label in accordance with MIL-L-1497 that complies with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. Each tube and paper label shall be labeled with the production date. The label shall show the following statement:

DO NOT EAT FOOD PRODUCT IF TUBE IS SWOLLEN.

3.7.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

3.8 Workmanship. The tubes shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

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4.2.1 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be examined for the defects specified in 4.5.3. The presence of any defect shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Certification. Certificates of compliance may be submitted for the requirements for interior coating and sealing compound. Nonconformance to the requirements shall be cause for rejection of the lot.

4.5.1.2 Copper sulfate testing. The empty tubes shall be tested for metal exposure as specified in 4.6.1. The lot size shall be expressed in units of tubes. The sample unit shall be one empty tube. Thirteen tubes shall be tested from each lot. Any evidence of black spots shall be a major defect and shall be cause for rejection of the lot.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the processing and sealing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed tube inspection. The filled and sealed tubes shall be inspected for tube defects as indicated in table I. The lot size shall be expressed in units of tubes. The sample unit shall be one filled and sealed tube. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5.

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TABLE I. Filled and sealed tube defects

Category		Defect
Critical	Major	
101		Hole or tear
	102	Not material specified
	103	Screw cap missing
	104	Loose screw cap
	105	Tube distorted (other than paneling) 1/
	106	Presence of sharp creases or indentation
	107	Three or more tubes are unclean
	108	Dimensions not as specified (see 3.2.2)
	109	Exterior coating missing, when specified
	110	Neck of tube not as specified
	111	Not folded saddle back or crimped
	112	Label missing
	113	Nomenclature missing, incorrect, or illegible
	114	Labeling, other than nomenclature, missing, incorrect, or illegible

1/ Paneling is the collapsing of a tube caused by a vacuum resulting from the cooling of a heated product. It commonly occurs at the crimped end of the tube.

4.5.4 Commercial sterility testing. The filled and sealed tubes shall be tested for commercial sterility. The sample unit shall be one filled and sealed tube. The sample size shall be ten filled, sealed, and thermoprocessed tubes selected from each batch in the lot. Tubes shall be selected from different areas within the batch. They shall be incubated at 72 to 86°F for ten days. Any evidence of swelling shall be considered a critical defect and shall be cause for rejection of the lot.

4.5.5 Examination for floaters. The filled, sealed, and pasteurized tubes shall be examined on a 100 percent basis in accordance with 4.6.2. Tubes that do not sink to the bottom of the tank shall be considered unacceptable (this applies to partial floaters also) and shall be excluded from delivery.

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4.5.6 Unit pack examination. Paperboard cartons shall be examined in accordance with the appendix to PPP-B-566. In addition, the following defects shall be included in the table of examination:

Major: Not packed as specified
Minor: Separators not as specified

4.5.7 Shipping container examination. Shipping containers shall be examined in accordance with the appendix to PPP-B-636. In addition, the following defects shall be included in the table of examination:

Major defects:	Marking missing, incorrect, or illegible
	More than one type or style
Minor defects:	Arrangement not as specified
	Not packed as specified
	Height of separators not as specified
	Not snug-fitting

4.6 Methods of inspection.

4.6.1 Copper sulfate test.

4.6.1.1 Materials. The copper sulfate solution shall consist of the following:

a. Copper sulfate cystals - 20% by weight.
b. 37% (concentrated) hydrochloric acid - 10% by weight.
c. Distilled water - 70% by weight.

CAUTION: Care should be exercised in handling the copper sulfate solution because it is quite corrosive. The solution should not be permitted to splash on the skin or clothing; neither should metal vessels be employed in its transfer or storage.

4.6.1.2 Procedure.

a. Remove the shoulder of the tube with shears.

b. Thoroughly rinse all tubes to be tested to ensure the removal of any foreign material.

c. Thoroughly drain, but do not dry, the test tubes.

d. Arrange tubes to be copper sulfated upright on a porcelain or soapstone sink top in rows of eight, this being the maximum number that can be handled at one time.

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e. Set timer for 60-second intervals.

f. Fill exposed shoulder to completely submerge shoulder with the copper sulfate solution. As the first tube is filled, start timer. Note: The copper sulfate solution can be used for repeat testing until it changes from blue to green in color, the green indicating considerable metal pickup.

g. After 60 seconds, empty tubes in the same order in which they were filled. The copper sulfate may be poured into other tubes to be tested or into storage carboys.

h. Thoroughly rinse the copper sulfated tubes with running tap water to remove all copper sulfate solution. Three rinses are generally adequate.

i. Allow tubes to air dry.

j. Any metal exposed area by evidence of black spots shall be a failure.

4.6.2 Floater test. Place filled, sealed, and cooled tubes in a flat bottom container filled with cold tap water to a depth of approximately two feet. Keep the tubes being tested separate from each other to allow them to move freely in the water. Tubes that sink to the bottom of the tank shall be considered to have air satisfactorily removed from the product. Tubes that do not sink to the bottom of the tank shall be considered unacceptable (total or partial floaters are included).

5. PACKAGING

This section is not applicable to this document.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type of product to be tube packed (see 3.2.1).
c. Type of exterior tube body coating required (see 3.2.3).
d. Type of processing required (see 3.4).
e. Type and class of unit load required (see 3.6).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

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6.3 Use information. Product documents available at the time of approval of this document were as follows:

Document No.	Product
MIL-P-44318(GL)	Peaches, Semi-Solid, Tube Pack
MIL-P-44319(GL)	Pears, Semi-Solid, Tube Pack
MIL-A-44321(GL)	Applesauce, Semi-Solid, Tube Pack
MIL-A-44322(GL)	Apple Pie, Semi-Solid, Tube Pack

6.4 Subject term (key word) listing.

Packaging
Semi-solid foods
Tube pack

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

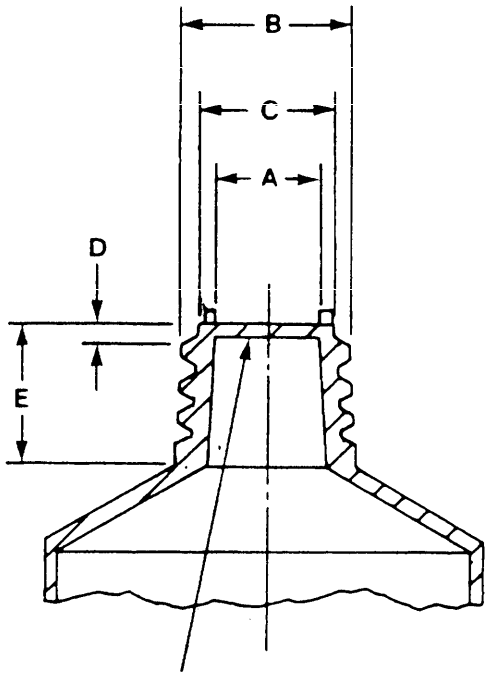
Army - MD, TS
Navy - MC
DP-SS

Preparing activity:

Army - GL

Project No. 8940-A593

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DIAPHRAGM THICKNESS: 0.004 ± 0.002 INCH
DIAPHRAGM DEPTH: 0.100 ± 0.016 INCH

SYM	DESCRIPTION	DIMENSION (inches)	TOLERANCE (inches)
A	ORIFICE DIAMETER	0.340	± 0.010
B	NECK DIAMETER	0.560	± 0.010
C	PILOT DIAMETER	0.480	± 0.010
D	PILOT LENGTH	0.032	± 0.016
E	NECK LENGTH	0.312	± 0.016

FIGURE 1. Tube dimensions.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-44320(CL)	Packaging And Processing Of Foods In Collapsible Containers (Tubes)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK (P), 1 Apr 82

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